



Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

Kinematics 7 Supplement: the four forms of

$$v^2 - u^2 = 2 a s \text{ — ANSWERS}$$

1. Order of operations — BIDMAS

Always remember BIDMAS!

	Stands for	What to do	Example
B	Brackets	calculate inside brackets first	$(3 + 2) \times 4 = 5 \times 4 = 20$
I	Indices (powers)	then do exponents	$3 + 4^2 = 3 + 16 = 19$
DM	Division, Multiplication	then $\div$ or $\times$, left to right	$12 \div 3 \times 2 = 4 \times 2 = 8$
AS	Addition, Subtraction	then $+$ or $-$, left to right	$10 - 3 + 2 = 7 + 2 = 9$

More Tips:

- **FRACTIONS** - consider these like Brackets. Do the top calculations, then the bottom calculations, then divide the whole thing.
- **SQUARE ROOT** - A square root sign works like brackets. Work out everything under the $\sqrt{\quad}$, then calculate the $\sqrt{\quad}$ after.

2. SUVAT — what each letter means

Symbol	Name	Unit	Unit in words
s	displacement (distance)	m	metres
u	initial velocity	m/s	metres per second
v	final velocity	m/s	metres per second
a	acceleration	m/s ²	metres per second squared
t	time	s	seconds

A. Find the distance, s

Worked example A — find the distance

A bus speeds up from 4.0 m/s to 12 m/s with an acceleration of 2.0 m/s². Calculate the distance it travels while speeding up.

What the question tells you

Initial velocity = u = 4.0 m/s

Final velocity = v = 12 m/s

Acceleration = a = 2.0 m/s²

Distance = s = ? *the one the question asks for*

Time = t = *not in the question: circle t in the Givens and cross it out*

1. Givens

$$s = ?$$

$$u = 4.0 \text{ m/s}$$

$$v = 12 \text{ m/s}$$

$$a = 2.0 \text{ m/s}^2$$

$$\cancel{t} =$$

2. Equation

$$v^2 - u^2 = 2 a s$$

3. Substitute

$$12^2 - 4.0^2 = 2 \times 2.0 \times s$$

4. Rearrange

$$s = \frac{12^2 - 4.0^2}{2 \times 2.0} \quad \text{form A}$$

5. Calculate

$$v^2 = 12^2 = 144$$

$$u^2 = 4.0^2 = 16$$

$$v^2 - u^2 = 144 - 16 = 128$$

$$2 \times a = 2 \times 2.0 = 4.0$$

$$s = 128 \div 4.0 = \mathbf{32 \text{ m}}$$

Check: $v^2 - u^2 = 128$ and $2 a s = 2 \times 2.0 \times 32 = 128$. Same number.

Now try. Fill in the Givens, circle and cross out t , then work down the ladder, one line at a time.

A1. A car speeds up from 10 m/s to 20 m/s with an acceleration of 3.0 m/s². Calculate the distance.

Givens

$$s = ?$$

$$u = 10 \text{ m/s}$$

$$v = 20 \text{ m/s}$$

$$a = 3.0 \text{ m/s}^2$$

$$\cancel{t} =$$

Form: $s = \frac{v^2 - u^2}{2 a}$

$$v^2 = 20^2 = 400$$

$$u^2 = 10^2 = 100$$

$$v^2 - u^2 = 400 - 100 = 300$$

$$2 \times a = 2 \times 3.0 = 6.0$$

$$s = 300 \div 6.0 = \mathbf{50 \text{ m}}$$

Check: $v^2 - u^2 = 300$

$2 a s = 300$

B. Find the acceleration, a

Worked example B — find the acceleration

A sprinter speeds up from 3.0 m/s to 9.0 m/s over a distance of 24 m. Calculate the sprinter's acceleration.

1. Givens

$$s = 24 \text{ m}$$

$$u = 3.0 \text{ m/s}$$

$$v = 9.0 \text{ m/s}$$

$$a = ?$$

$$\cancel{\text{X}} =$$

2. Equation $v^2 - u^2 = 2 a s$

3. Substitute $9.0^2 - 3.0^2 = 2 \times a \times 24$

4. Rearrange $a = \frac{9.0^2 - 3.0^2}{2 \times 24}$ *form B*

5. Calculate $v^2 = 9.0^2 = 81$

$$u^2 = 3.0^2 = 9.0$$

$$v^2 - u^2 = 81 - 9.0 = 72$$

$$2 \times s = 2 \times 24 = 48$$

$$a = 72 \div 48 = \mathbf{1.5 \text{ m/s}^2}$$

Check: $v^2 - u^2 = 72$ and $2 a s = 2 \times 1.5 \times 24 = 72$. Same number.

Now try. In form B the bottom line is $2 \times s$.

B1. A bike slows from 8.0 m/s to a stop over 16 m. Calculate the acceleration. Slowing down gives a negative acceleration, so your answer has a minus sign.

Givens

$$s = \mathbf{16 \text{ m}}$$

$$u = \mathbf{8.0 \text{ m/s}}$$

$$v = \mathbf{0}$$

$$a = ?$$

$$\cancel{\text{X}} =$$

Form: $a = \frac{v^2 - u^2}{2 s}$

$$v^2 = \mathbf{0^2 = 0}$$

$$u^2 = \mathbf{8.0^2 = 64}$$

$$v^2 - u^2 = \mathbf{0 - 64 = -64}$$

$$2 \times s = \mathbf{2 \times 16 = 32}$$

$$a = \mathbf{-64 \div 32 = -2.0 \text{ m/s}^2}$$

Check: $v^2 - u^2 = \mathbf{-64}$ $2 a s = \mathbf{-64}$

C. Find the final velocity, v

Worked example C — find the final velocity

A scooter moving at 9.0 m/s accelerates at 2.0 m/s² over a distance of 10 m. Calculate its final velocity.

1. Givens

$$s = 10 \text{ m}$$

$$u = 9.0 \text{ m/s}$$

$$v = ?$$

$$a = 2.0 \text{ m/s}^2$$

$$\cancel{\text{X}} =$$

2. Equation

$$v^2 - u^2 = 2 a s$$

3. Substitute

$$v^2 - 9.0^2 = 2 \times 2.0 \times 10$$

4. Rearrange

$$v = \sqrt{9.0^2 + 2 \times 2.0 \times 10} \text{ form C}$$

5. Calculate

$$u^2 = 9.0^2 = 81$$

$$2 a s = 2 \times 2.0 \times 10 = 40$$

$$v^2 = 81 + 40 = 121$$

$$v = \sqrt{121} = \mathbf{11 \text{ m/s}}$$

Check: $v^2 - u^2 = 121 - 81 = 40$ and $2 a s = 40$. Same number.

Now try. Add first, and press $\sqrt{}$ last.

C1. A cyclist moving at 3.0 m/s accelerates at 3.0 m/s² over 12 m. Calculate the final velocity.

Givens

$$s = \mathbf{12 \text{ m}}$$

$$u = \mathbf{3.0 \text{ m/s}}$$

$$v = ?$$

$$a = \mathbf{3.0 \text{ m/s}^2}$$

$$\cancel{\text{X}} =$$

Form: $v = \sqrt{u^2 + 2 a s}$

$$u^2 = \mathbf{3.0^2 = 9.0}$$

$$2 a s = \mathbf{2 \times 3.0 \times 12 = 72}$$

$$v^2 = \mathbf{9.0 + 72 = 81}$$

$$v = \mathbf{\sqrt{81} = 9.0 \text{ m/s}}$$

Check: $v^2 - u^2 = \mathbf{72}$ $2 a s = \mathbf{72}$

D. Find the initial velocity, u

Worked example D — find the initial velocity

A motorbike accelerates at 3.0 m/s^2 over 20 m and reaches 13 m/s . Calculate how fast it was going at the start.

1. Givens

$$s = 20 \text{ m}$$

$$u = ?$$

$$v = 13 \text{ m/s}$$

$$a = 3.0 \text{ m/s}^2$$

$$\cancel{\text{X}} =$$

2. Equation

$$v^2 - u^2 = 2 a s$$

3. Substitute

$$13^2 - u^2 = 2 \times 3.0 \times 20$$

4. Rearrange

$$u = \sqrt{13^2 - 2 \times 3.0 \times 20} \quad \text{form D}$$

5. Calculate

$$v^2 = 13^2 = 169$$

$$2 a s = 2 \times 3.0 \times 20 = 120$$

$$u^2 = 169 - 120 = 49$$

$$u = \sqrt{49} = \mathbf{7.0 \text{ m/s}}$$

Check: $v^2 - u^2 = 169 - 49 = 120$ and $2 a s = 120$. Same number.

Now try. Subtract first, and press $\sqrt{}$ last.

D1. A car accelerates at 2.0 m/s^2 over 7.0 m and reaches 8.0 m/s . Calculate its initial velocity.

Givens

$$s = 7.0 \text{ m}$$

$$u = ?$$

$$v = 8.0 \text{ m/s}$$

$$a = 2.0 \text{ m/s}^2$$

$$\cancel{\text{X}} =$$

Form: $u = \sqrt{v^2 - 2 a s}$

$$v^2 = 8.0^2 = 64$$

$$2 a s = 2 \times 2.0 \times 7.0 = 28$$

$$u^2 = 64 - 28 = 36$$

$$u = \sqrt{36} = \mathbf{6.0 \text{ m/s}}$$

Check: $v^2 - u^2 = 28$ $2 a s = 28$

E. Calculator round — one calculator question; answer to 2 significant figures

Same steps: one calculator line at a time. Round only on the last line, and check with the number before you rounded.

E1. A van moving at 4.0 m/s accelerates at 1.2 m/s² over 50 m. Calculate the final velocity.

Givens

$$s = 50 \text{ m}$$

$$u = 4.0 \text{ m/s}$$

$$v = ?$$

$$a = 1.2 \text{ m/s}^2$$

$$\cancel{\text{X}} =$$

$$\text{Form: } v = \sqrt{u^2 + 2 a s}$$

$$u^2 = 4.0^2 = 16$$

$$2 a s = 2 \times 1.2 \times 50 = 120$$

$$v^2 = 16 + 120 = 136$$

$$v = \sqrt{136} = 11.66 = \mathbf{12 \text{ m/s}}$$

Check: $v^2 - u^2 = 120$ $2 a s = 120$